Fault Detection in a Launch Vehicle Pitch Control System Using IMU Measurements

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Abstract

This report investigates control stability and fault detection in a single-axis launch vehicle pitch rate control system. The vehicle dynamics are modeled using linearized rigid-body rotational dynamics with aerodynamic damping, while actuator behavior is represented by a first-order lag. A proportional controller is designed using phase margin criteria to ensure stable closed-loop operation.

A step gyro bias fault is introduced in the angular rate measurement to study fault propagation in the feedback loop. A residual-based detection scheme using threshold logic and moving-average filtering is implemented to identify sensor faults. Simulation results show that actuator lag reduces stability margins, while increasing measurement noise leads to larger fault detection delays. Despite these effects, the detection algorithm identifies faults before significant instability develops.

1 Introduction

Attitude control is essential for maintaining launch vehicle stability during flight. Angular rate measurements obtained from inertial measurement units (IMUs) are commonly used in feedback loops to regulate vehicle motion. However, sensor faults such as gyro bias can introduce errors in the measured angular rate and propagate through the control system.

In closed-loop systems, a bias in the gyro measurement can cause the controller to generate unintended control inputs, leading to physical drift in the vehicle motion. In addition, actuator dynamics introduce phase lag in the feedback loop, reducing stability margins and affecting control performance.

This work analyzes the interaction between actuator dynamics, gyro bias faults, and fault detection in a single-axis pitch rate control system. A residual-based detection method is implemented to identify sensor bias faults, and the influence of measurement noise on detection delay is evaluated through simulation.

2 System Modeling

2.1 Modeling Assumptions

The study considers pitch rate regulation using direct gyro feedback. To obtain a control-oriented representation, the system is linearized under small-angle motion and modeled

as a rigid body with constant inertia J. Aerodynamic effects are approximated as rate-proportional damping, and actuator dynamics are modeled as a first-order lag. Higher-order nonlinearities and additional vehicle dynamics are neglected.

2.2 Rigid-Body Pitch Dynamics

Rotational motion about the pitch axis is governed by

$$\sum M = J\dot{\omega} \tag{1}$$

where J is the pitch moment of inertia and ω is the angular rate. Considering control torque $M_{control}$ and aerodynamic damping $M_{aero} = D\omega$, the governing equation becomes

$$J\dot{\omega} = M_{control} - D\omega \tag{2}$$

Rearranging,

$$\dot{\omega} = -\frac{D}{J}\omega + \frac{1}{J}M_{control} \tag{3}$$

Taking the Laplace transform (zero initial conditions),

$$G_{rigid}(s) = \frac{\Omega(s)}{M(s)} = \frac{1}{Js + D} \tag{4}$$

2.3 Actuator Dynamics

In practice, actuators cannot respond instantaneously to command inputs due to mechanical and control dynamics.

$$\tau\dot{u}(t) + u(t) = u_c(t) \tag{5}$$

where $u_c(t)$ is the commanded input, $u(t)$ is the actual actuator output, and τ is the actuator time constant.

Taking the Laplace transform (zero initial conditions),

$$(\tau s + 1)U(s) = U_c(s) \tag{6}$$

Thus, the actuator transfer function becomes

$$G_{act}(s) = \frac{U(s)}{U_c(s)} = \frac{1}{\tau s + 1} \tag{7}$$

The parameter τ represents actuator responsiveness, with larger values corresponding to slower response and increased phase lag in the feedback loop.

2.4 Combined Plant

The overall plant is obtained by cascading actuator and rigid-body dynamics:

$$G_{plant}(s) = \frac{1}{(\tau s + 1)(Js + D)} \tag{8}$$

The system contains two poles: a rigid-body pole and an actuator pole. The actuator pole directly affects achievable bandwidth and robustness.

3 Stability Analysis and Controller Design

3.1 Open-Loop Stability Analysis

The open-loop plant is given by

$$G_{plant}(s) = \frac{1}{(\tau s + 1)(Js + D)} \tag{9}$$

The system contains two poles: a rigid-body pole at $-D/J$ and an actuator pole at $-1/\tau$. While both poles lie in the left-half plane and are therefore stable, the actuator pole introduces additional phase lag in the feedback loop.

To evaluate robustness, frequency-domain analysis was performed using Bode plots. Increasing the actuator time constant τ moves the actuator pole closer to the origin, reducing phase margin and limiting achievable control bandwidth.

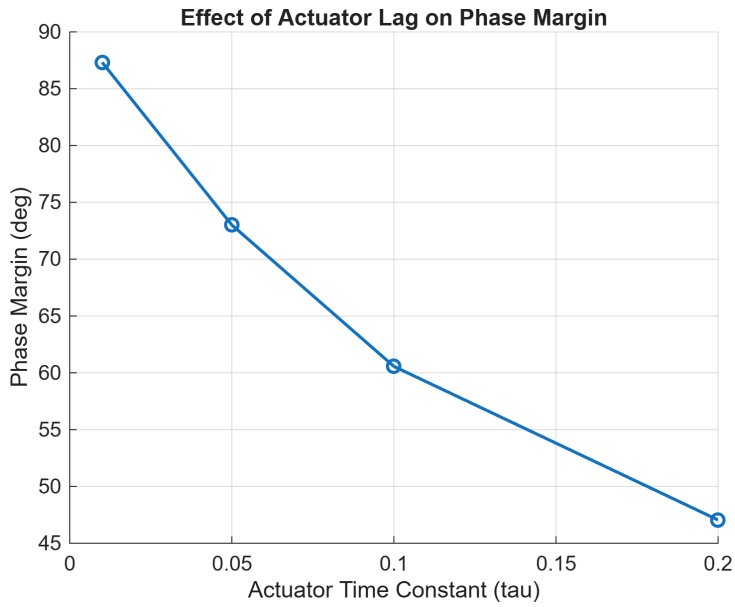

Figure 1: Effect of actuator time constant on phase margin. Larger actuator lag reduces stability margin.

3.2 Proportional Controller Design

A proportional controller of the form

$$C(s) = K_p \tag{10}$$

was selected to regulate the vehicle pitch rate.The objective of the controller is to regulate the angular rate to zero while maintaining sufficient stability margin.

A full PID controller was not adopted to isolate the interaction between actuator dynamics and fault propagation. Since pitch rate is directly regulated, integral action was not required, and derivative action was avoided because it amplifies measurement noise. The controller gain K_p was tuned using phase margin criteria to maintain a phase margin greater than 50°.

Through iterative tuning, the selected gain was $K_p = 35{,}000$, which yielded:

- Phase Margin $\approx 60°$
- Overshoot $\approx 9\%$
- Settling Time ≈ 0.7 s

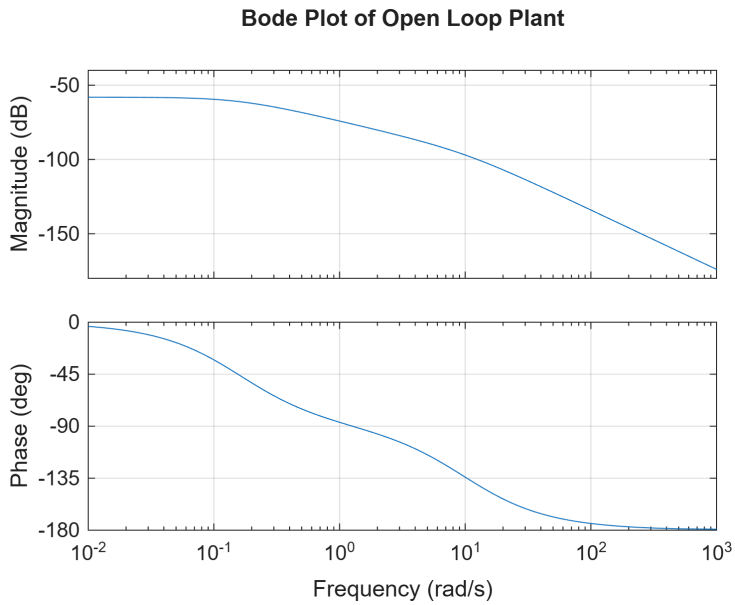

Figure 2: Open-loop Bode plot with proportional controller ($K_p = 35{,}000$). The phase margin of approximately 60° satisfies the robustness requirement of PM $\geq 50°$.

Figure 2 confirms that the selected gain achieves the desired stability margin. The gain crossover frequency occurs near 6 rad/s, with a phase margin of approximately 60°, providing adequate robustness while maintaining acceptable transient performance.

3.3 Effect of Gain on Stability Margin

Increasing the controller gain K_p improves response speed but reduces phase margin. For example, increasing the gain to $K_p = 45{,}000$ reduces the phase margin to approximately 42°. The system remains stable but with reduced robustness to actuator lag and modeling

uncertainties.

4 Fault Modeling and Injection

4.1 Gyro Bias Fault Model

The measured angular rate is modeled as

$$\omega_{meas}(t) = \omega_{true}(t) + b(t) + n(t) \tag{11}$$

where $b(t)$ represents gyro bias and $n(t)$ represents zero-mean Gaussian measurement noise.

A step bias fault is introduced at $t = 5$ s,

$$b(t) = \begin{cases} 0, & t < 5 \\ b_0, & t \geq 5 \end{cases} \tag{12}$$

where b_0 denotes the bias magnitude.In this study, the bias magnitude was set to $b_0 = 0.02$ rad/s.

4.2 Fault Propagation Mechanism

In the closed-loop system, the controller uses ω_{meas} instead of ω_{true}. Therefore, the control input becomes

$$u(t) = K_p \left(\omega_{ref} - \omega_{meas}(t)\right) \tag{13}$$

For rate regulation, $\omega_{ref} = 0$. When a bias fault occurs, the controller interprets the bias as a real rotational error and generates a compensating torque, causing the true angular rate to drift until a new equilibrium is reached. Thus, sensor bias propagates into vehicle motion through the feedback loop.

4.3 Noise Modeling

Measurement noise is modeled as zero-mean Gaussian noise,

$$n(t) \sim \mathcal{N}(0, \sigma^2) \tag{14}$$

where σ represents noise standard deviation. Different noise levels are evaluated to study detection sensitivity and robustness.

4.4 Observed Fault Behavior

Simulation results show that immediately after bias injection, control torque experiences a transient spike as the controller attempts to compensate for the perceived rate error. The true angular rate drifts toward a value approximately equal and opposite to the bias, restoring the measured error toward zero.

The rate of drift depends on controller gain, actuator dynamics, and phase margin.

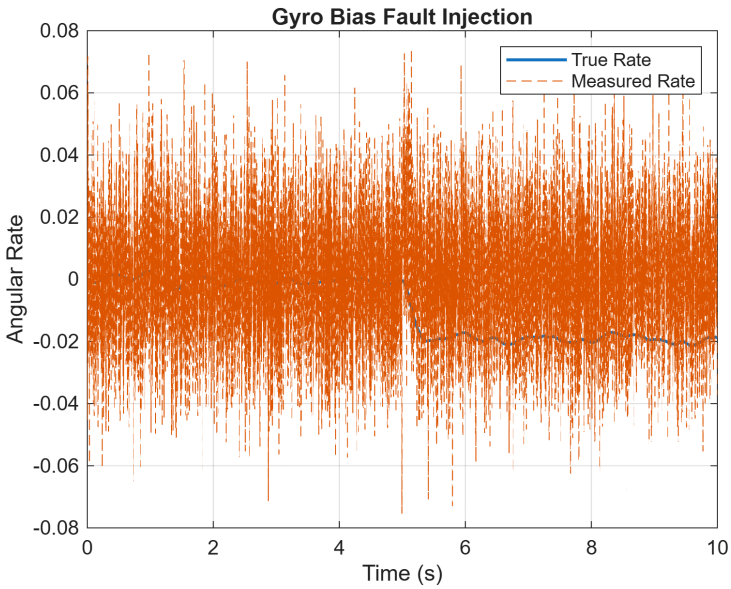

Figure 3: Pitch rate response under step gyro bias injection at $t = 5$ s. The measured rate exhibits an offset while the true rate drifts toward a new equilibrium.

5 Residual-Based Fault Detection

5.1 Residual Definition

To detect gyro bias faults, a residual signal is defined as the difference between measured and true angular rate,

$$r(t) = \omega_{meas}(t) - \omega_{true}(t) \tag{15}$$

In simulation, the true angular rate is available from the plant model, allowing direct computation of the residual.Under nominal conditions, the residual consists mainly of measurement noise.. When a bias fault occurs, the residual shifts approximately by the bias magnitude.

5.2 Threshold Logic and Persistence Condition

A threshold-based detection scheme is implemented such that a fault is declared when

$$|r(t)| > \gamma \tag{16}$$

for a continuous duration greater than a persistence window T_w.. The threshold γ is selected relative to the expected noise amplitude to balance sensitivity and false alarm rejection.In this study, the threshold was set to $\gamma = 0.015$ rad/s based on the expected noise amplitude to avoid false alarms while maintaining sensitivity to the injected bias fault. A persistence window of $T_w = 0.2$ s was selected to prevent spurious detections caused by short-duration noise spikes.

5.3 Residual Filtering

To reduce the effect of high-frequency measurement noise, a moving-average filter (moving mean) is applied to the residual before threshold evaluation. While filtering reduces noise variance, it introduces a small detection delay.

5.4 Detection Results

Figure 4 shows the filtered residual signal for a representative case with bias injection at $t = 5$ s. The residual remains near zero under nominal conditions and crosses the detection threshold shortly after fault injection.

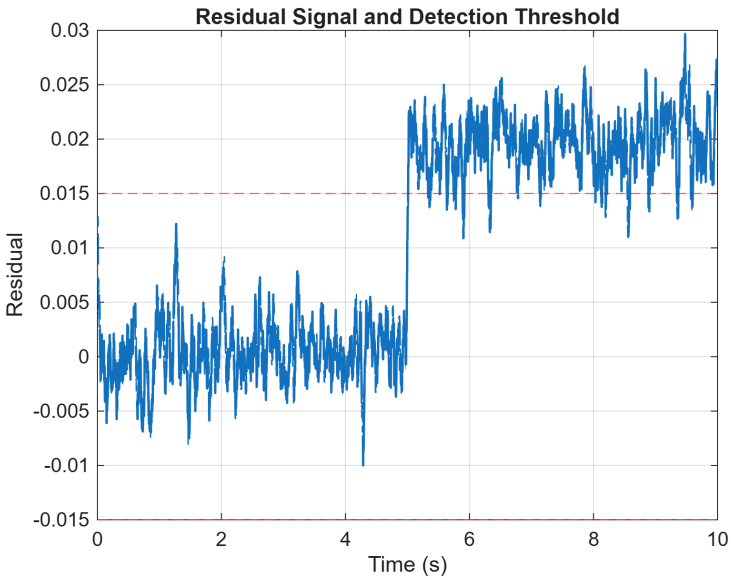

Figure 4: Filtered residual signal with threshold bounds. A bias fault is injected at $t = 5$ s.

Detection delay increases with measurement noise amplitude. Under low noise, detection occurs within a few milliseconds. At higher noise levels, delay increases but remains small relative to the dominant system time constant, preventing significant instability growth.

6 Comparative Results and Robustness Analysis

To evaluate system robustness, multiple scenarios were simulated by varying actuator dynamics, controller gain, and measurement noise. The objective was to analyze how

these parameters affect stability margin, fault detection delay, and physical drift at the moment of detection.

6.1 Test Scenarios

The following parameters were varied:

- Actuator time constant τ
- Controller gain K_p
- Noise standard deviation σ

For each case, phase margin, detection delay, and true angular rate at detection were recorded.

6.2 Quantitative Comparison

Table 1: Robustness analysis under varying actuator dynamics, controller gain, and measurement noise

τ	K_p	σ	PM (°)	Detection Delay (s)	Drift at Detection (rad/s)
0.1	35,000	0.005	60	0.0128	-0.01925
0.1	35,000	0.02	60	0.0570	-0.01966
0.1	35,000	0.05	60	1.6588	-0.01857
0.2	35,000	0.02	47	0.0872	-0.01944
0.2	45,000	0.02	42	0.0504	-0.01933

Table 1 summarizes representative simulation scenarios illustrating the influence of noise, actuator lag, and controller gain on system robustness and fault detection performance.

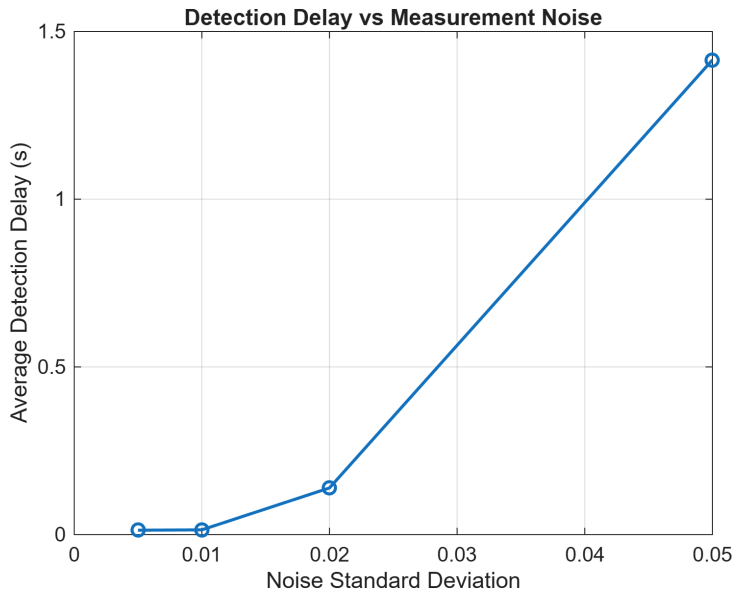

Figure 5: Detection delay as a function of measurement noise standard deviation.

Figure 5 shows the relationship between detection delay and measurement noise standard deviation. Detection delay increases as noise amplitude grows because the separation between the bias signal and measurement noise becomes smaller. As a result, the residual requires a longer persistence duration before exceeding the detection threshold.

6.3 Discussion of Results

The results show that actuator lag primarily affects stability margin but does not significantly alter detection time. Detection delay is dominated by measurement noise and filtering parameters.

As noise amplitude increases, detection delay grows due to reduced signal-to-noise separation in the residual. For very high noise levels ($\sigma = 0.05$), the detection delay reaches approximately 1.6 s, which is longer than the nominal closed-loop settling time. Larger delays allow greater physical drift before the fault is detected. However, in all tested scenarios, the fault was detected before significant instability developed.

7 Discussion and Conclusion

The results show that actuator dynamics and measurement noise influence system robustness in different ways. Increasing the actuator time constant introduces additional phase lag, reducing phase margin but having little effect on fault detection timing.

Measurement noise primarily affects the residual evaluation. As noise amplitude increases, separation between nominal fluctuations and bias-induced shifts decreases, resulting in longer detection delays and greater physical drift before fault declaration.

Despite these effects, the proposed residual-based detection scheme successfully identified the injected gyro bias fault in all tested scenarios. Detection occurred before significant instability developed, indicating that the selected controller and detection logic provide adequate robustness under moderate noise and actuator lag conditions.

Overall, the study demonstrates the trade-off between detection sensitivity and robustness, highlighting the importance of selecting appropriate thresholds relative to system dynamics and measurement noise levels.

8 Future Work

The present study considered a simplified single-axis pitch rate control system with threshold-based fault detection. Future work could extend this analysis by incorporating attitude estimation methods such as Kalman filtering, applying observer-based fault detection techniques, and modeling multi-axis vehicle dynamics. Additional improvements may include adaptive threshold selection and more detailed actuator and aerodynamic models to better represent practical aerospace control systems.